

Program: Diploma in Civil & Environmental Engineering	
Course Code: 6352C	Course Title: Climate Change and Sustainability
Semester: 6	Credits: 4
Course Category: Open Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the science behind climate change, its impacts on the built environment, and the strategies for promoting sustainability in civil engineering projects.
- To explain climate change mitigation and adaptation, sustainable construction practices, and the role of civil engineers in creating a resilient future.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Sustainability	4353	Sustainable Engineering	4
Basic knowledge of Environmental Engineering		Environmental Engineering	5

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Explain the science of climate change, including causes, impacts, and global efforts to mitigate it.	15	Understanding
CO2	Describe sustainability principles to civil engineering projects and construction practices.	15	Understanding
CO3	Explain methods to reduce the carbon footprint in civil engineering design and construction.	14	Understanding
CO4	Describe strategies for climate change adaptation in infrastructure projects.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	3	-	-
CO2	3	-	-	-	3	-	-
CO3	3	-	-	-	3	-	-
CO4	3	-	-	-	3	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the science of climate change, including causes, impacts, and global efforts to mitigate it.		
M1.01	Describe the fundamental concepts of climate change, including the greenhouse effect, global warming, and the scientific basis behind climate change.	7	Understanding
M1.02	Identify the natural and anthropogenic causes of climate change, including the role of greenhouse gases, deforestation, and industrial emissions.	5	Understanding
M1.03	Explain significance of international agreements like the Kyoto Protocol and the Paris Agreement in global efforts to combat climate change.	3	Understanding
Contents: Basic concepts of climate change: Definition, global warming, and the greenhouse effect. Scientific evidence of climate change: Global temperature rise, ice melting, sea level rise, and extreme weather events. Causes of climate change: Natural vs. anthropogenic causes, greenhouse gases, deforestation, and urbanization. International climate change agreements: The Kyoto Protocol, Paris Agreement, and United Nations Framework Convention on Climate Change (UNFCCC)			
CO2	Describe sustainability principles to civil engineering projects and construction practices.		

M2.01	Identify the potential impacts of climate change on civil infrastructure such as transportation, buildings, and energy systems.	7	Understanding
M2.02	Describe the social and environmental consequences of climate change, including migration, displacement, and the loss of biodiversity.	5	Understanding
M2.03	Explain the principles of resilience and adaptive capacity and apply them in civil engineering to design infrastructure that can withstand climate change impacts.	3	Understanding
	Series Test -I	1	

Contents:

Impacts of climate change on infrastructure: Flooding, droughts, heatwaves, storm surges, and rising sea levels.

Vulnerability of civil engineering systems: Transportation, water supply, waste management, energy systems, and buildings.

Impact on communities and ecosystems: Migration, displacement, loss of biodiversity, and social consequences.

Case studies: Examples of climate change impacts on infrastructure around the world.

CO3	Explain methods to reduce the carbon footprint in civil engineering design and construction.		
M3.01	Explain the strategies for mitigating climate change through the reduction of greenhouse gas emissions, energy efficiency, and renewable energy adoption.	6	Understanding
M3.02	Describe techniques to reduce the carbon footprint in civil engineering design and construction, focusing on life cycle assessments and carbon-neutral projects.	5	Understanding
M3.03	Explain adaptation strategies for civil engineering infrastructure, such as flood-resistant buildings, water conservation, and stormwater management systems.	3	Understanding

Contents:

Mitigation strategies: Reducing greenhouse gas emissions through energy efficiency, renewable energy, and carbon capture technologies.

Adaptation strategies: Designing infrastructure to withstand climate change effects (e.g., flood-resistant buildings, stormwater management).

Carbon footprint reduction: Life cycle assessment (LCA), embodied carbon, and strategies for reducing emissions in construction.

Urban planning for climate resilience: Green urban planning, sustainable transportation,

and low-carbon cities.			
CO4	Describe strategies for climate change adaptation in infrastructure projects.		
M4.01	Explain the core principles of green building, including sustainable site development, water and energy efficiency, and material conservation.	4	Understanding
M4.02	Describe global green building certification systems such as LEED and IGBC to assess the sustainability of construction projects.	6	Understanding
M4.03	Explain sustainable construction practices, including waste reduction, reuse of materials, and recycling in construction projects.	4	Understanding
	Series Test – II	1	
Contents: Green building concepts: Sustainable site development, water efficiency, energy efficiency, and material conservation. Green building certification systems: LEED (Leadership in Energy and Environmental Design), IGBC (Indian Green Building Council), and other global standards. Designing for sustainability: Incorporating passive design, energy-efficient HVAC systems, and renewable energy sources in buildings. Sustainable construction practices: Waste reduction, reuse, and recycling in construction projects.			

Text / Reference:

T/R	Book Title/Author
T1	Sustainable Development in Civil Engineering by I. D. Singh
T2	Climate Change and Sustainable Development: A Synthesis of the Key Issues by J. P. Newell
R1	Climate Change and Its Impact on the Built Environment by S. K. Garg
R2	Green Architecture: Advanced Technologies and Materials by Osman Attmann

Online resources:

Sl.No.	Website Link
1	https://www.nptel.ac.in/